THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The difference of the squares of two consecutive natural numbers is 59. The numbers are: 1

(a) 29 and 30 **(b)** 29 and 31 **(c)** 30 and 31 **(d)** 28 and 29

2. The ones digit of 7^{23} is: 1

(a) 5 **(b)** 7 **(c)** 3 **(d)** 9

3. The number of 1s in the binary form of 77 is: 1

(a) 5 **(b)** 4 **(c)** 3 **(d)** 6

4. The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 100^\circ$, then $\angle OAB$ is: 1

(a) 65° **(b)** 50° **(c)** 100° **(d)** 40°

5. The smallest 3-digit palindrome divisible by 3 is: 1

(a) 121 **(b)** 111 **(c)** 101 **(d)** 114

6. The number of digits in the square root of a 5-digit perfect square is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

7. If $3^x = 243$, then 3^{x-2} equals: 1

(a) 2187 **(b)** 241 **(c)** 9 **(d)** 27

8. The Roman numeral for 1994 is: 1

(a) MCMXCIII **(b)** MCMXCIV **(c)** MMIV **(d)** MCMXCV

9. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = (3x + 10)^\circ$ and $\angle D = (4x + 30)^\circ$. The measure of $\angle C$ is: 1

(a) 110° **(b)** 145° **(c)** 70° **(d)** 130°

10. If the number 6a2b is divisible by 90, then $a + b$ equals: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

11. $\sqrt[3]{-729}$ is: 1

(a) -9 **(b)** 9 **(c)** -10 **(d)** 10

12. The value of $(2^{-1} + 3^{-1})^{-1}$ is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{6}{5}$ **(c)** 6 **(d)** $\frac{5}{6}$

13. The number that comes immediately after 10111_2 in the binary system is: 1

(a) 11001_2 **(b)** 10110_2 **(c)** 101110_2 **(d)** 11000_2

14. The side of a rhombus is $(3x + 2)$ cm and its perimeter is 80 cm. The value of x is: 1

(a) 6 **(b)** 8 **(c)** 5 **(d)** 7

15. The number of steps the Kaprekar routine needs to reach 6174 starting from 2111 is: 1

(a) 4 **(b)** 7 **(c)** 6 **(d)** 5

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The difference between the squares of two consecutive numbers is 75. Find the numbers. 2

17. Write $(3 \times 10^5) + (4 \times 10^4)$ in scientific notation. 2

18. Write 90 in binary and count the number of 1s in it. 2

19. The angles of a quadrilateral are in the ratio 2:5:6:7. Find the largest angle and say whether it is acute, right or obtuse. 2

20. How many multiples of 11 lie between 150 and 800 (both included)? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Add 343_5 and 212_5 (numbers in base 5) and give the answer in base 5. 3

22. The angles of a quadrilateral are $x°$, $(2x - 10)°$, $(3x - 20)°$ and $(4x - 30)°$. Find all four angles. 3

23. Find the greatest 4-digit number divisible by 6, 8 and 12. 3

24. A rectangular hall 36 m by 48 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

OR

Find the value of $\dfrac{2^6 + 2^5}{2^4 \times (2 + 1)}$. 3

25. How many different numbers can be written using at most 8 digits in base 2? What is the largest of them in decimal? 3

OR

One angle of a parallelogram is 2 times its adjacent angle. Find all the angles of the parallelogram. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the digits a and b if the number 4a3b is divisible by 45. List all possibilities. 5

OR

Find the smallest cube number that is divisible by each of 4, 6 and 9. 5

27. Body A has mass 6×10^{24} kg and body B has mass 3×10^{22} kg. Which is heavier? How many times as heavy is it? Find the total mass in scientific notation. 5

OR

Subtract $1222_5 - 341_5$ in base 5. Verify your answer in decimal. 5

END OF QUESTION PAPER — **Total: 50 Marks**